## SPECIFIC AIMS

Immune checkpoint inhibitors (ICIs) have transformed the treatment of advanced cancers – including non-small cell lung cancer (NSCLC), bladder cancer, kidney cancer (renal cell carcinoma), and melanoma – leading to durable responses and improved survival but often requiring prolonged treatment that carries ongoing risks of immune-related adverse events (irAEs) and other burdens (e.g., treatment costs, lost time at home). Many patients remain alive and progression-free after prolonged ICI therapy. However, a fundamental clinical question persists: **how long should ICI therapy continue?** Most pivotal randomized controlled trials (RCTs) assigned a fixed treatment duration of two years, largely for practical rather than biological reasons. Patients who reach this milestone without progression or significant toxicity must decide whether to discontinue therapy to reduce cumulative irAEs and treatment burden or continue therapy to potentially improve survival. Clinical practice varies widely, and guidelines provide limited direction. Existing observational studies are limited by methodological challenges, including immortal time bias, time-varying confounding, and difficulty distinguishing elective discontinuation from discontinuation due to disease progression or toxicity. In addition, some RCTs addressing this question have faced feasibility challenges due to poor accrual.

**Our overall objective** is to generate rigorous evidence on the comparative effectiveness of discontinuing versus continuing ICI therapy at the two-year duration milestone among patients with advanced cancers who are alive and progression-free. **Our central hypothesis** is that discontinuation of ICI therapy will be non-inferior to continuation with respect to overall survival while reducing cumulative irAE risk, with heterogeneity across tumor types and patient subgroups. A modest difference in survival (5% absolute risk difference at two years after the continuation/discontinuation decision point) is clinically acceptable if accompanied by meaningful reductions in toxicity. We will emulate a target trial using real-world oncology electronic health record (EHR) data from the nationwide Flatiron Health database and apply advanced causal inference methods (clone-censor-weighting with inverse probability weighting) to address immortal time bias and selection bias and generate rigorous real-world evidence on treatment duration.

Our specific aims are:

**Aim 1. Among people with advanced cancer who are alive and progression-free after two years of ICI therapy, estimate the difference in subsequent two-year overall survival probability comparing those who discontinue ICI therapy at two years versus those who continue.** We will assess non-inferiority using a prespecified margin informed by clinical considerations.

*Hypothesis 1: Discontinuation will be non-inferior to continuation with respect to overall survival.*

**Aim 2. Among people with advanced cancer who are alive and progression-free after two years of ICI therapy, estimate the difference in the cumulative incidence of irAEs comparing those who discontinue ICI therapy at two years versus those who continue.**

*Hypothesis 2: Continued therapy beyond two years will be associated with higher irAE cumulative incidence.*

**Aim 3. Develop individualized risk prediction models to inform decisions about discontinuation versus continuation of ICI therapy.** We will develop and internally validate clinical risk prediction models to estimate individualized risks of survival and irAEs under each strategy. Using regularized regression to flexibly integrate interactions between treatment duration and tumor types and patient characteristics (e.g., age, body mass index, performance status, PD-L1 expression level, and tumor mutational burden status, among others), we will develop a model that provides personalized predictions of benefits and harms of discontinuation at two years versus continuation. The final model will be implemented as a prototype web-based R Shiny app to support individualized risk prediction.

*Hypothesis 3: Clinical risk prediction models for overall survival and irAEs will provide accurate and well-calibrated estimates of two-year risks under each strategy.*

This study will generate rigorous real-world evidence on whether discontinuation of ICI therapy after two years duration preserves survival benefits while reducing toxicity. Findings will inform decision-making, reduce unnecessary treatment, and improve the value of cancer care. More broadly, this work will demonstrate how causal inference methods applied to real-world data can address clinically actionable non-inferiority questions that are not amenable to investigation in RCTs.

## A. SIGNIFICANCE

This proposed study addresses a critical and unresolved clinical effectiveness question faced by a growing population of patients with advanced (i.e., metastatic or stage IV) cancer treated with immune checkpoint inhibitors (ICIs). Although ICI treatment has dramatically improved outcomes for certain advanced cancers, continued use of this treatment comes with risks of immune-related adverse events (irAEs) and ongoing treatment burdens (e.g., treatment costs, lost time at home). Together with their providers, patients whose cancer remains under control on therapy must decide how long to continue ICI treatment. *Making decisions about discontinuing versus continuing treatment requires new evidence. Generating rigorous real-world evidence, informed by patient, clinician, and other stakeholder priorities, is the focus of this application.*

**A.1. Patients with advanced cancers treated with immune checkpoint inhibitors represent a large and growing population facing an important unresolved treatment decision regarding therapy duration.** ICIs have fundamentally transformed the treatment of advanced cancers, including non-small cell lung cancer (NSCLC), urothelial (bladder) cancer, renal cell carcinoma (RCC, i.e., kidney cancer), and melanoma. Collectively, these cancers account for more than 350,000 new diagnoses and over 200,000 deaths annually in the United States.[1] For patients with advanced disease, ICIs have resulted in unprecedented improvements in survival, with randomized controlled trials (RCTs) showing 30-50% reductions in all-cause mortality compared with prior standards of care.[2–5] As a result, ICIs are now widely used as first-line therapy for many patients with advanced solid tumors. Recent estimates from our work suggest that between one-quarter and one-half of U.S. patients with advanced lung, bladder, or kidney cancer receive ICI therapy.[6,7]

With improved survival, a growing population of patients receiving ICI therapy now remains alive without cancer progression for extended periods. Traditionally, patients with advanced cancer remain on a treatment until their cancer progresses or they have unmanageable side effects. However, a fundamental question remains unanswered: *how long should ICI therapy continue?* Nearly all pivotal RCTs assigned a fixed treatment duration of two years. This duration was based on trial design practicality rather than biology, leaving major uncertainty about the benefits and harms of shorter or longer treatment courses. As a result, current clinical practice reflects convention, not evidence. The National Comprehensive Cancer Network (NCCN) treatment guidelines recommend two years of ICI treatment for advanced lung cancer, if tolerated, but no definitive guidance exists for melanoma, kidney, or bladder cancer. In the absence of evidence-based guidance, real-world practice varies widely across clinicians and health systems.[8]

This uncertainty has major implications for patients and populations. Continued therapy exposes patients to cumulative ICI-related toxicities including pneumonitis, colitis, endocrinopathies, and adrenal insufficiency – some of which may be permanent and require lifelong management,[9] and others may not appear until therapy is discontinued. Prolonged therapy also imposes substantial treatment burden including ongoing infusions, monitoring, travel, missed work, and psychological strain, as well as financial strains on patients and healthcare systems.[10,11] Conversely, discontinuing therapy may generate anxiety about recurrence or concern about stopping effective therapy (**Box 1**). Importantly, ICIs can have durable biological effects after discontinuation because of their long half-lives and sustained immune activation.[12] This suggests that continued treatment may not confer additional survival benefit for patients who are progression-free at two years. At a population level, variation in practice may result in inconsistent care and avoidable irAEs without clear evidence of additional benefit.

> ***Box 1.*** *"I have been on an ICI for just about two years now and my doctor says I could stop, but I'm afraid to. It would help to know whether people like me who stop are more likely to have progression than those who don't stop, or if they have fewer new side effects."*
>
> -Jane Perlmutter, Patient Advocate

**A.2. Existing trial and observational data provide important insights but do not provide evidence to guide the decision of whether to stop or continue ICI therapy at the two-year treatment decision point.**

*Evidence from Randomized Controlled Trials.* Long-term follow up from pivotal trials, including KEYNOTE-006[2] in melanoma and KEYNOTE-189[5] in NSCLC, demonstrate durable survival among patients treated with ICIs, including those who discontinue at the protocol-defined two-year time-point. However, these trials did not randomize patients who were progression-free at the two-year milestone to continue versus discontinue. Thus, they do not provide information on a critical decision that patients face, which is the focus of this application.

*Feasibility Challenges of Randomized Controlled Trials of ICI Continuation vs. Discontinuation.* RCTs evaluating treatment duration have proven difficult to conduct. The CheckMate-153 trial[13] remains the only RCT comparing fixed duration versus continuous ICI in NSCLC and showed worse outcomes among patients randomized at one year to stop versus continue ICI. Notably, the trial did not evaluate discontinuation at two years. Similarly, the UK-based DANTE trial attempted to randomize patients with metastatic melanoma who were progression-free after one year of ICI therapy to continuation versus discontinuation but closed early due to poor accrual.[14] Such recruitment barriers likely reflect patient reluctance to discontinue a therapy perceived to be life-prolonging. As a result, large RCTs evaluating optimal ICI duration are unlikely to be completed in the near future.

*Findings from Observational Studies of Elective ICI Discontinuation.* Outside of RCTs, a growing body of evidence from observational studies has attempted to evaluate the relationship between ICI treatment duration and outcomes. Real-world analyses using electronic health records (EHR) data, including prior work co-authored by MPI Dr. Mamtani (Sun et al[8]) have suggested that subsequent survival outcomes for patients with NSCLC may be similar whether ICIs are stopped at two years or continued indefinitely. Systematic reviews and retrospective cohorts have reached similar conclusions in NSCLC,[15,16] although results vary across other tumor types including melanoma, kidney, and bladder cancer.[17]

Retrospective cohorts across melanoma, kidney cancer, and NSCLC suggest that many patients maintain durable disease control after stopping ICIs, particularly those achieving deep responses.[18,19] However, these studies include highly selected populations enriched for responders to treatment, further limiting generalizability. These studies do not emulate a clearly defined treatment decision at a specific time point; instead, they describe outcomes among patients who discontinue therapy at variable times and for different reasons. As a result, they cannot estimate the causal effect of a defined discontinuation strategy (e.g., stopping at two years) compared with continued therapy (until progression or irAE). A recent study from France using target trial emulation in a melanoma cohort has begun to address some of these challenges, but this study focused on earlier treatment timepoints and had very few patients who could contribute to analyses of the two-year decision faced by many patients receiving ICI today.[20]

*Methodological Limitations of Observational Studies of ICI Continuation vs. Discontinuation.* Despite these insights, observational studies of ICI continuation vs. discontinuation share several important risks of bias. First, if studies classify patients according to their observed treatment duration and initiate follow-up at treatment start, they are effectively treating duration as a fixed (time invariant) exposure. This introduces immortal time bias, whereby patients receiving longer treatment appear to have better outcomes simply because they survived long enough to remain on therapy. Second, some studies combine patients who discontinue treatment due to progression or irAEs with those who electively discontinue therapy while clinically stable. This approach conflates distinct clinical scenarios and limits relevance of results to the key question faced by patients who are progression-free at two years. Our proposed project will apply rigorous methods to avoid these sources of bias.

*Persistent Evidence Gap and Need for Rigorous Comparative Effectiveness Research.* Expert reviews and commentaries have repeatedly highlighted the uncertainty surrounding ICI treatment duration and called for rigorous comparative effectiveness research using real-world data and modern causal inference methods.[21–23] These commentaries and others have emphasized the biologic plausibility of durable immune activation after ICI treatment cessation,[21–24] alongside the increasing risk of irAEs and treatment burden associated with prolonged therapy. Yet no prior well-powered study has emulated the clinical decision faced by patients and clinicians at the two-year decision point while minimizing confounding, selection bias, and immortal time bias.

*Thus, an important evidence gap remains: existing studies have not rigorously evaluated whether patients who remain progression-free after two years of ICI therapy should discontinue or continue treatment. Generating high-quality evidence to inform this decision is essential for supporting shared decision-making among patients, clinicians, and health systems seeking to balance durable cancer control with irAEs and treatment burden.*

**A.3. Patients and clinicians face a critical decision regarding whether to stop or continue ICI therapy at the two-year treatment decision point.** The proposed study addresses the following question:

*For patients with advanced NSCLC, bladder cancer, kidney cancer, or melanoma who reach the two-year treatment milestone on ICI treatment while remaining progression-free, should ICI therapy be discontinued or continued?*

This question represents a common and high-stakes clinical decision. At the two-year treatment decision point, oncologists and patients must decide whether to:

1. Discontinue ICI therapy, consistent with clinical trial precedent, or

2. Continue ICI therapy, in the hope of further improving survival.

Both strategies are used in routine practice. In a preliminary analysis of the Flatiron Health oncology EHR database, we found that only 23% of patients reaching the two-year treatment decision point discontinued at that time. Yet neither strategy is strongly supported by comparative data. The decision, therefore, represents a state of true clinical uncertainty. Continued therapy may offer potential survival benefit but increase cumulative irAEs and treatment burden. Discontinuation may reduce treatment-related harms but may generate anxiety about the risk of relapse.

Limited qualitative research suggests that patients and caregivers have diverse perspectives on treatment duration. In one qualitative study of melanoma and kidney cancer survivors, some patients reported that continued treatment visits provided reassurance and support, whereas stopping therapy could feel like abandonment.[25] Conversely, others emphasized the burdens of ongoing treatment, including irAEs, travel, and interference with work or family responsibilities, and expressed preference for shorter treatment courses. Many emphasized that treatment decisions should be personalized and guided by evidence.

Patients and clinicians confirmed that this decision is common, emotionally complex, and inadequately informed by evidence. Through two clinician interviews and focus groups with six patients/caregivers, stakeholders emphasized the lack of clear evidence to guide discontinuing versus continuing ICIs, the importance of balancing survival with the burdens of therapy (irAEs), and the desire for personalized decision support (preliminary qualitative work conducted in preparation for this study). Patients expressed a strong desire for individualized, data-driven guidance rather than a "one-size-fits-all" approach. Clinicians emphasized the need for rigorous evidence to support shared decision-making conversations.

As the number of patients reaching the two-year treatment milestone continues to grow, closing this evidence gap becomes increasingly critical for patients, caregivers, clinicians, health systems, and payers.

**A.4. The proposed study compares two real-world management strategies that patients and clinicians frequently face in routine oncology care.** The interventions under study correspond directly to competing clinical strategies: Discontinue ICI therapy at two years, consistent with the duration used in pivotal trials versus continue ICI therapy beyond two years until progression or irAE. The outcomes selected, overall survival and irAEs, reflect endpoints that patients and clinicians have identified as meaningful: length of life and risk of treatment-related harm. By focusing exclusively on patients who are progression-free at two years, the study isolates the moment when this decision arises in practice, maximizing clinical relevance and patient-centeredness.

*Why This Study is Needed Now*. Because RCTs of duration are unlikely to resolve this question soon and existing observational studies are methodologically limited, rigorous real-world comparative effectiveness research is urgently needed. By leveraging large-scale EHR data and modern causal inference methods, this study will estimate the benefits and harms of discontinuing versus continuing ICIs at the two-year milestone. These findings will directly inform shared decision-making and help reduce variations in care for a growing population of cancer survivors.

## B. INNOVATION

This study is innovative because it combines advanced causal inference methods, rich real-world data, and individualized risk modeling to address an important question in oncology that has not been resolved by RCTs.

- First, we apply a target trial emulation framework – an established causal inference approach – to a clinical question where it has not previously been implemented: the decision to discontinue versus continue ICI therapy at two years. Prior observational studies of ICI duration have been limited by immortal time bias, selection bias, and poorly defined treatment strategies. By explicitly emulating the clinical decision point and using a clone-censor-weight approach with inverse probability weighting, we will address these limitations and generate estimates that aim to approximate those from a randomized trial.
- Second, we use the Flatiron Health oncology EHR database, which captures treatment patterns, real-world progression, and clinically abstracted irAEs. This allows us to evaluate both survival and toxicity in the same study. These data are key to unbiased estimation for the clinical question of interest and are not available at

scale in most other real-world data sources.

- Third, we move beyond population-level treatment effects to develop individualized risk prediction models. These models will estimate both survival and irAE risk under each treatment strategy. We will use regularized regression with interaction terms to produce stable, personalized estimates. To our knowledge, these models will be the first focused on individualized risk prediction for advanced cancer patients who have been on treatment and progression-free for an extended period of time.
- Finally, we will translate these findings into a prototype web-based individualized risk prediction tool (an R Shiny app) that generates personalized risk estimates under each treatment strategy, laying the groundwork for future assessment and evaluation in clinical settings.

## C. APPROACH

**C.1. Study team and preliminary data.** Our study team has extensive experience studying cancer outcomes and emulating target trials with real-world data. Specifically, our team has used Flatiron Health EHR data to construct treatment-defined cancer cohorts and estimate real-world treatment effectiveness, comparing ICI vs non-ICI therapies in over 34,000 cancer patients using methodological approaches similar to those proposed in this study[7], with findings that inform treatment decisions in populations underrepresented in clinical trials. This work demonstrates our ability to rigorously define real-world treatment strategies, mitigate confounding, and generate meaningful comparative effectiveness evidence using large-scale EHR data. Our team has experience with a wide variety of real-world data sources including data from integrated health care delivery systems (Chubak and Hubbard),[26–31] SEER-Medicare (Hubbard, Lund),[32–34] and Flatiron Health (Hubbard, Mamtani).[7,35–39] Our team has replicated clinical trial eligibility within Flatiron's longitudinal cancer cohorts[37,40] and has implemented target trial emulation investigating the comparative effectiveness of alternative cancer treatments.[40]

In a prior publication on which MPI Dr. Mamtani was an author, we did not observe a notable increase in discontinuation of ICI therapy (in the absence of progression or death) at two years compared to other timepoints.[8] In fact, only 23% of patients on ICI therapy at two years discontinued at that point in time. There was no obvious subsequent time point at which discontinuation was especially common (**Figure 1**). Our preliminary data from Flatiron suggest that among patients who were progression-free and on ICI at 24 months, about 70% (bladder 70%, kidney 69%, NSCLC 74%, melanoma 68%) were still progression-free and taking ICI therapy three months later. These findings show that, in the absence of progression, *ICI therapy use for more than two years is widespread*.

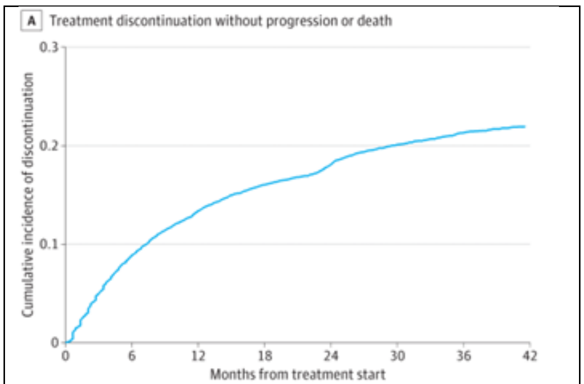


**Figure 1.** Treatment discontinuation in the absence of progression or death, starting at treatment initiation (Figure 3A from Sun et al[8]).

We also have experience with developing guidance for target trial emulation. Dr. Chubak was part of a team that developed guidance for how to emulate target trials using case-control data.[41] Dr. Lund has work applying target trial emulation approaches using observational data in cancer research based on claims data and registry data,[42–44] and an ongoing NIH-funded R01 which also applies these methods for lung cancer screening (R01CA277756). Dr. Lund also led an international working group of experts from the International Society for Pharmacoepidemiology (ISPE) to develop a primer for clinical oncology audiences on target trial emulation, and specifically, the application of the clone-censor-weight approach for cancer research.[45] This primer includes the development of visual tools for communicating key aspects of the study design and its application.

**C.2. Conceptual Framework/Causal Model.** Unlike traditional chemotherapy, which works by directly killing cancer cells, ICI therapy works indirectly by promoting an immune response to tumor cells.[46] Some tumor cells express molecules that can bind to receptors (e.g., programmed cell death-1 or PD-1) on immune system cells (e.g., T cells) and block them from killing tumor cells. Essentially, ICIs block tumor cells from flipping immune system cells from "on" to "off". Because they target the immune system, the effect of ICIs can persist, for some patients, after they stop treatment. RCTs have shown that benefits of randomization to two years of ICI treatment are maintained into long-term follow-up. However, evidence suggests that at least some minimum duration of use is necessary. For example, the Checkmate 153 RCT found that progression-free survival and

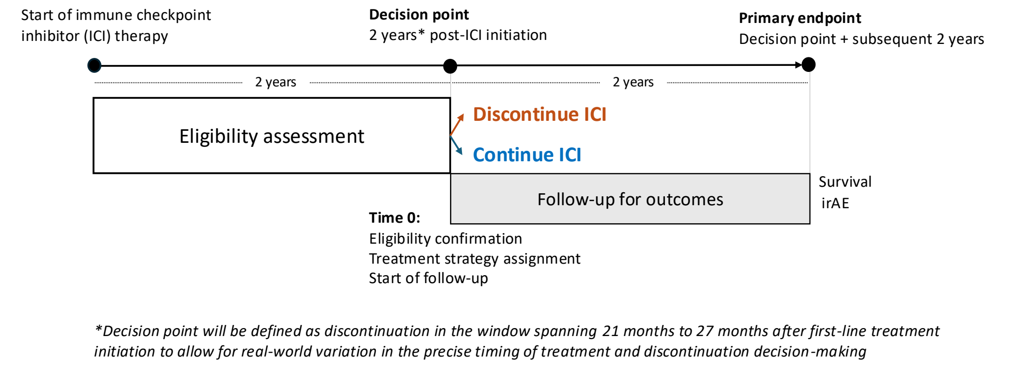


**Figure 2.** Conceptual illustration of key clinical time points based on the conceptual model for the treatment decision design.

overall survival were worse among patients who used ICI therapy for one year and were then randomized to discontinue, compared to those randomized to continue treatment.[13] And, in a French target trial emulation, Amiot et al. found that advanced melanoma patients who discontinued ICI treatment after six months had worse outcomes than those who continued.[20] On the other hand, data also show that irAEs can occur even after a patient has been on treatment for years.[47]

The tension between *durability of treatment response* and *ongoing risk of irAEs* makes decisions about stopping versus continuing treatment salient for patients on ICIs and their providers. Our proposed study is based on a *treatment decision design approach*. This approach anchors epidemiologic studies on "the time when a treatment decision is made",[48] comparing outcomes among groups who have made different choices. Patients and providers must, over time, weigh the risks of continuing ICI treatment with the risks of discontinuing ICI treatment **(Figure 2)**. The decision to continue might come with increased risks of irAEs and related deaths, while the decision to discontinue treatment may come with an increased risk of disease progression and therefore cancer-specific mortality.

**C.3. Study Population and Research Setting.** This retrospective observational study will use real-world data from the nationwide Flatiron Health oncology EHR database. We selected this data source due to its careful curation and its suitability for research based on population size, availability of necessary data elements, and representativeness of real-world oncology practice. Taken together with our team's familiarity and prior experience in working with the data, these factors make Flatiron an excellent – and potentially the only – existing data source in which to address questions about ICI duration. Flatiron Health data have been used to study cancer treatment in routine clinical practice, helping to address concerns about the limited generalizability of RCTs to real-world populations and care settings. [e.g.,35,49–52]

These data are derived from the EHRs of >280 community and academic oncology practices across the US. The network represents a geographically diverse, longitudinal cohort of 2.2 million patients, integrated into a single data warehouse via a cloud-based platform. Flatiron Health provides granular patient-level data, including data on progression and select irAEs critical for this research, not otherwise captured in traditional cancer datasets. Data elements include patient demographics, dates of advanced cancer diagnoses, treatment dates, type of systemic therapy, Eastern Cooperative Oncology Group (ECOG) performance status, comorbidity, smoking history, laboratory results (liver/renal function, tumor biomarkers), medications, and validated endpoints (mortality, real-world progression, select irAEs).[53–55] A key strength of the Flatiron database is its ability to capture select, clinically validated irAEs through technology-enabled chart abstraction, a feature that is rarely available at scale in other EHR-derived or registry-based oncology data sources. Flatiron Health data are curated using technology-enabled chart abstraction via machine learning and trained abstractors (e.g., oncologists and tumor registrars)[56] and key data elements have been validated.[57–59] Patients in Flatiron are generally demographically representative of US cancer registry populations, except that there is a higher proportion of cases with advanced disease (an inclusion criterion for this study) in Flatiron.[60] We will use data from all sites represented in the Flatiron Health database.

The proposed study will focus on adults (age 18+ years) diagnosed with advanced bladder, kidney, melanoma, or non-small cell lung cancer between 2015-2024. We selected these cancer types because ICI therapy is indicated as a first-line treatment, there is clear evidence of clinical uncertainty regarding the harms and benefits of discontinuing treatment at two years for these cancers, and Flatiron provides sufficiently rich, longitudinal data to rigorously evaluate these questions. The years selected for inclusion align with availability of ICI as first-line treatment for advanced cancer while also ensuring the potential for at least two years of ICI treatment (i.e., a key inclusion criterion) prior to the decision timepoint to be investigated. We will exclude NSCLC patients whose tumors have sensitizing mutations of EGFR or ALK that confer worse outcomes with

Results stratified by these factors

ICI therapy. Key planned subgroup analyses include stratification by cancer type, age group, body mass index (BMI), ECOG performance status, programmed death ligand (PD-L1) status, tumor mutational burden (TMB) status, and calendar period of ICI initiation. Preliminary data on patient demographic and clinical characteristics for patients currently in the Flatiron Health oncology EHR database meeting inclusion criteria are presented in **Table 1**.

Our study team has successfully conducted multiple prior studies using the Flatiron Health database[7,8,35,37,38,40,61,62] and has established an effective process for rapid dataset curation and analysis. Data will be accessed at the University of Pennsylvania via standard approval and delivery processes. High data recency (i.e., minimal lag between care delivery and data availability) and rapid data delivery are additional strengths of the Flatiron Health database that support the success of this project.

**C.4. Study Design.** We seek to investigate whether, among patients who are alive and progression-free at the two-year milestone, discontinuation of ICI therapy is non-inferior to continuation with respect to overall survival, while also quantifying differences in irAE risk. Discontinuation of ICI therapy may reduce cumulative irAEs and treatment burden. A modest decrement in survival may be clinically acceptable if accompanied by meaningful reductions in toxicity. Thus, a noninferiority framework is appropriate. We defined a noninferiority margin of no more than 5% absolute difference in 2-year overall survival based on clinical judgement. This is at the lower end of margins used in noninferiority trials in oncology.[63] The study population comprises adults with advanced non-small cell lung cancer, bladder cancer, kidney cancer, or melanoma who initiated first-line ICI therapy and remain progression-free and on treatment at two years. We compare two clinically relevant strategies: (1) discontinuation of ICI therapy at 2 years and (2) continuation of ICI therapy beyond 2 years until disease progression or development of clinically significant irAEs.

In sensitivity analyses, we will vary the discontinuation/continuation decision point from two years to earlier time points (e.g., 18 months). Amiot et al. used this approach of varying decision points for ICI discontinuation in melanoma.[20] While their two-year analysis (based on a very small sample size of only 14 discontinuers and

**Table 1**. Distribution of patient characteristics overall and by cancer type for patients initiating first line treatment with ICIs who are progression-free and on ICI treatment at 2 years in the Flatiron Health oncology EHR database.

| | Overall N (%) | NSCLC N (%) | Bladder Cancer N (%) | Melanoma N (%) | Kidney Cancer N (%) |
|---|---|---|---|---|---|
| **Total** | 7837 | 5980 | 624 | 644 | 589 |
| **Age at diagnosis (years)** | | | | | |
| **19-34** | 13 (0.2) | 9 (0.2) | 0 (0.0) | 0 (0.0) | 4 (0.7) |
| **35-49** | 306 (3.9) | 161 (2.7) | 3 (0.4) | 15 (2.4) | 127 (21.6) |
| **50-64** | 1903 (24.3) | 1601 (26.8) | 36 (5.7) | 57 (8.9) | 209 (35.6) |
| **65-74** | 2578 (32.9) | 2180 (36.5) | 73 (11.7) | 166 (25.8) | 159 (27.1) |
| **75-84** | 2277 (29.1) | 1886 (31.5) | 122 (19.5) | 185 (28.8) | 84 (14.4) |
| **85+** | 368 (4.7) | 143 (2.4) | 17 (2.7) | 204 (31.6) | 4 (0.7) |
| **Sex** | | | | | |
| **Female** | 3324 (42.4) | 2779 (46.5) | 171 (27.4) | 207 (32.1) | 167 (28.4) |
| **Male** | 4513 (57.5) | 3201 (53.5) | 453 (72.6) | 437 (67.9) | 422 (71.6) |
| **Race and ethnicity** | | | | | |
| **Hispanic or Latino** | 61 (0.8) | 8 (0.1) | 0 (0.0) | 0 (0.0) | 53 (9.0) |
| **Non-Hispanic Asian** | 119 (1.5) | 101 (1.7) | 8 (1.2) | 1 (0.2) | 9 (1.6) |
| **Non-Hispanic Black** | 713 (9.1) | 641 (10.7) | 28 (4.5) | 3 (0.5) | 40 (6.8) |
| **Non-Hispanic White** | 5402 (68.9) | 4046 (67.7) | 434 (69.5) | 537 (83.4) | 385 (65.5) |
| **Other Race** | 511 (6.5) | 391 (6.5) | 57 (9.2) | 32 (4.9) | 30 (5.1) |
| **Unknown** | 1030 (13.1) | 792 (13.2) | 97 (15.5) | 70 (10.9) | 71 (12.0) |
| **PD-L1 % staining** | | | | | |
| **1-49%** | 1671 (21.3) | 1531 (25.6) | 61 (9.8) | 57 (8.9) | 22 (3.7) |
| **<1%** | 1411 (18.0) | 1265 (21.2) | 44 (7.1) | 58 (9.1) | 43 (7.4) |
| **≥50%** | 1777 (22.7) | 1741 (29.1) | 12 (1.9) | 18 (2.8) | 6 (0.9) |
| **No evidence of testing** | 2978 (38.0) | 1442 (24.1) | 507 (81.2) | 510 (79.2) | 518 (88.0) |

Will vary 2-year decision time-point via sensitivity analysis due to lack of consensus in results.

119 continuers) favored discontinuation, results were less clear for 12- and 18-month timepoints. In NSCLC, there is RCT evidence (Checkmate 153) that discontinuation at one year reduces overall survival and progression free survival among previously treated patients.[13] However, observational data in first-line NSCLC suggest that continuation beyond two years may not provide additional benefit.[8,15] There is a paucity of data on the effects of discontinuing versus continuing first-line ICI for metastatic kidney and bladder cancers.[17,64]

We will analyze a longitudinal cohort using a treatment decision design anchored at the two-year milestone. We will emulate a target trial using real-world data and apply a clone-censor-weight approach to compare the effects of these strategies. The primary estimand is the per-protocol effect of discontinuation versus continuation on overall survival (primary) and irAEs (secondary). Per protocol effects are feasible to estimate in observational studies and are the preferred estimand in noninferiority studies where intention to treat may be biased towards a finding of no difference (i.e., non-inferiority) between treatment arms.[65] The target trial emulation (TTE) framework proposes that observational studies should be designed and analyzed as similarly as possible to RCTs to estimate a well-specified causal effect with minimal bias. The researcher specifies a causal question of interest and outlines the RCT that they would ideally conduct if they could. This hypothetical RCT is the "target trial." The researcher describes key components for the target trial, including eligibility criteria, treatment strategies, assignment procedures, follow-up period, outcome, causal contrasts of interest, identification assumptions, and analysis plan.[66,67] Then, they specify the observational analog of these study elements, aligning the observational study design with RCT definitions as closely as possible. Advantages of this approach include precise specification of an estimand, improved alignment between observational study and RCT results, and reduced risk of immortal time bias and selection bias. Several studies that have used the TTE approach have shown good concordance with RCT results, even when more "traditional" observational approaches have not.[68,69] Consistent with guidelines for the design and reporting of TTEs,[67] we have pre-specified key design elements of the hypothetical target trial that we seek to emulate and its observational emulation in Flatiron EHR data (**Table 2**).

Defining estimands, and backing up this choice.

**C.5. Outcomes.** Overall survival, the key outcome in cancer treatment RCTs, is the primary outcome of Aim 1. irAEs, defined as any of the following incident irAEs available in Flatiron EHR data: colitis, pneumonitis, hypothyroidism, adrenal insufficiency, or hypophysitis, will be the primary outcome of Aim 2. Both survival and irAEs are readily available as structured, curated data elements in the Flatiron Health oncology EHR data. Cumulative incidence of outcomes following the two-year milestone was selected as the primary endpoint to capture long-term outcomes. Patients who are progression-free after two years of ICI therapy represent potential long-term responders, for whom understanding the consequences of stopping versus continuing treatment is of direct clinical relevance. A two-year outcome window following the two-year continuation/discontinuation decision point also ensures adequate follow-up time is available in Flatiron EHR data, balancing scientific interest in long-term outcomes with practical data availability.

**C.6. Statistical analysis plan.** We will conduct a longitudinal observational study of differences in the comparative effect of discontinuation at two years versus continued treatment among patients who are progression-free at the two-year milestone. Discontinuation at the "two-year milestone" will be defined as a gap of ≥90 days in treatment administration occurring in the window spanning 21 months to 27 months after first-line treatment initiation to allow for real-world variation in the precise timing of treatment and discontinuation decision-making. Our analysis targets the per protocol estimand by assigning patients to the treatment strategy that they are observed to receive, consistent with recommendations for non-inferiority studies.[65]

In observational studies, it is critical to align assessment of eligibility, start of follow-up, and assignment to treatment strategy to avoid selection bias and immortal time bias. Start of follow-up will therefore begin at 21 months after first-line treatment initiation. Eligibility criteria including being free from evidence of progression will also be assessed using data collected up to the start of the 21st month after treatment initiation. Since discontinuation anywhere in the window from 21 - 27 months after treatment initiation will be considered to be consistent with the "discontinue at two years" treatment strategy, treatment arm cannot be uniquely assigned at the start of follow-up. We will therefore use the clone-censor-weight approach to assign patients to treatment arms at baseline.[45,70,71] In this framework, data for patients who meet eligibility criteria at the start of follow-up (time 0) will be duplicated into two copies with one copy (or "clone") assigned to each of the two treatment strategies. These cloned observations will be followed for the outcome of interest until the earliest of death, end of follow-up, or deviation from their assigned treatment strategy. When a clone is observed to deviate from their assigned treatment strategy, they will be artificially censored. Observations assigned to the "discontinue" strategy will be considered to deviate from this strategy if the patient does not discontinue treatment by 27 months. Observations assigned to the "continue" strategy will be artificially censored at the time of

Administrative censoring defined by treatment arm, continue strategy vs discontinue strategy.

| **Table 2.** Specification of the hypothetical target trial and its observational emulation using Flatiron Health oncology EHR data. | | |
|---|---|---|
| | **Target trial specification** | **Emulation using EHR data** |
| **a) Eligibility criteria** | Patients with advanced NSCLC, kidney cancer, bladder cancer, or melanoma who were:<br>1. Treated with ICI as first-line therapy for advanced cancer<br>2. Alive and progression-free at two years after initiating ICI<br>3. Still receiving ICI treatment at two years after initiation<br>Progression-free status is assessed using standardized radiographic criteria (e.g. RECIST) at prespecified intervals.<br>Continued ICI use is verified at the two-year timepoint in clinical encounters. | Same as target trial, except:<br>• Advanced disease, histology, and line of therapy will be identified using Flatiron disease-specific extracted tables<br>• Progression-free status will be ascertained using clinician-documented progression in Flatiron's disease-specified progression tables<br>• ICI use at two years will be defined using Flatiron's drug and administration episode tables, requiring receipt of ICI within a pre-specified window (e.g., prior 90 days) |
| **b) Treatment strategies** | Two years after ICI initiation (time 0)<br>1. Continue ICI treatment until progression or grade 3 or higher irAEs, defined using The Common Terminology for Adverse Events (CTCAE) (v5.0)<br>2. Discontinue ICI treatment | Same as target trial, except:<br>• Treatment strategy is assessed using medication administration records.<br>• A grace period of 21–27 months is used to assess strategy adherence: the continue strategy is defined as the absence of a ≥90-day gap in ICI administration without concurrent irAE or progression through 27 months; the discontinue strategy is defined as the occurrence of such a gap within this window.<br>• irAEs and progression are collected from Flatiron's EHR-derived datasets |
| **c) Assignment procedures** | Randomized assignment to treatment strategies; participants and clinicians aware of assignment. | Non-randomized assignment based on observed treatment patterns. Treatment strategies emulated using a clone-censor-weight approach, assigning individuals to both strategies at time 0 (i.e., 21 months after ICI treatment initiation) and artificially censoring upon observed deviation from the assigned strategy. |
| **d) Follow-up** | Follow-up begins at randomization (two years after ICI initiation) and continues for two years or until death, loss to follow-up, or administrative censoring (end of study data). | Same as target trial, except follow-up will begin at start of grace period (i.e., 21 months after ICI initiation) to align with treatment strategy classification and avoid immortal time bias. |
| **e) Outcomes** | At two years after randomization<br>• Primary (Aim 1): Overall survival<br>• Primary (Aim 2): Immune-related adverse events (irAEs) assessed using CTCAE<br>• Secondary (Aim 2): Specific irAEs (colitis, pneumonitis, adrenal insufficiency, hypophysitis)<br>Effect measure: difference in cumulative incidence two years after assignment | Same as target trial, except:<br>• Overall survival will be obtained from Flatiron's vital status tables<br>• irAEs will be identified using Flatiron's EHR-derived adverse event documentation |
| **f) Causal contrasts** | Per protocol | Per protocol |
| **g) Identifying assumptions** | Consistency; positivity; conditional exchangeability with respect to adherence; no interference; non-informative censoring | Same as target trial, except exchangeability conditional on measured covariates (see Section D.1) |
| **h) Data analysis plan** | Cox proportional hazards model with inverse probability weighting for nonadherence and censoring | Pooled logistic regression with inverse probability of treatment and censoring weights. Account for grace period with clone-censor-weight approach. |
| *At start of follow-up, which will be two years after ICI initiation. | | |

discontinuation, if discontinuation occurs prior to the end of 27 months (see **Figure 3A-B**). For example, patient 5 in **Figure 3A** is observed to receive immunotherapy about every two months until they experience an irAE at month 33, discontinue in month 36 and die in month 47. Because at the start of follow-up in month 21, we cannot know whether they will discontinue or continue ICI treatment, we create two clones of patient 5 and assign a clone to each strategy (discontinue=pink; continue=blue). The discontinue (pink) clone 5 is artificially censored at month 27 because the patient has not discontinued, deviating from the clone's assigned strategy, while the continue (blue) clone 5 continues to be followed until they experience an irAE and ultimately death.

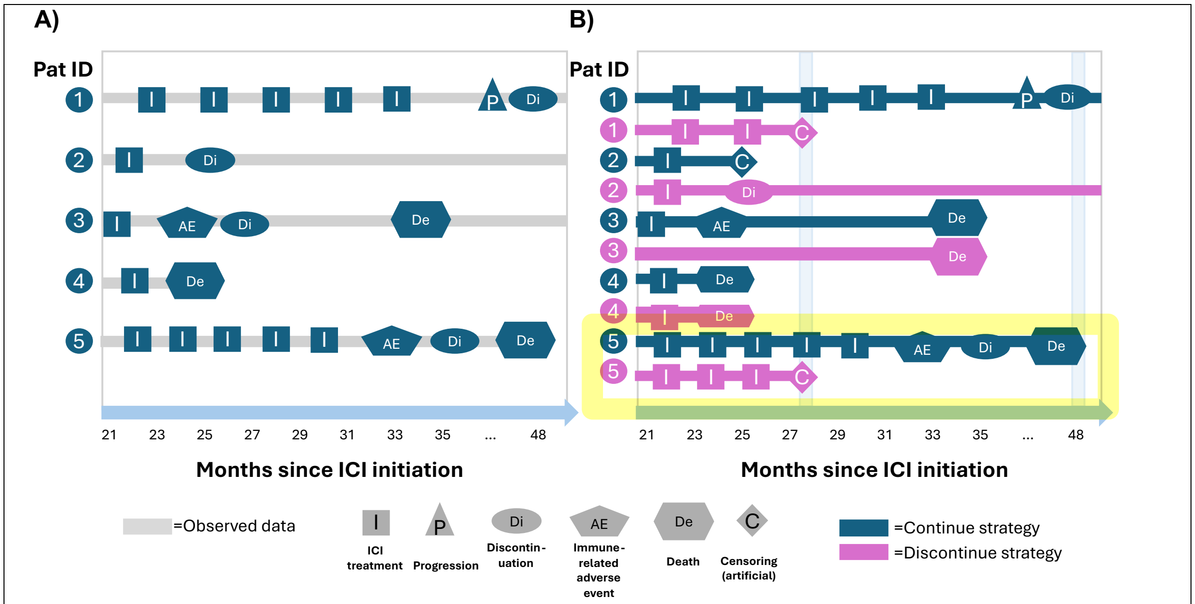


**Figure 3A-B.** Illustration of the clone-censor-weight approach applied to immune checkpoint inhibitor discontinuation (pink) versus continuation (blue). Panel A depicts the original data patient timeline for five hypothetical patients (all on ICI and free from progression at month 21), and Panel B depicts their corresponding cloned and censored patient timeline data.

***Aim 1.*** We will first conduct descriptive analyses to summarize patient characteristics, treatment patterns including frequency and timing of treatment discontinuation, and length of available follow-up. Baseline covariates will be summarized at the start of the 21-month decision window, and we will examine the distribution and timing of discontinuation, progression, and irAEs. We will also perform data quality assessments, including evaluation of missingness, internal consistency of key dates and events, and plausibility checks, to ensure the validity of analytic variables and inform model specification.

We will then estimate the per protocol effect of discontinuation at two years versus continued ICI treatment until progression or irAE on the primary outcome of overall survival using pooled logistic regression with inverse probability of censoring weights (IPCW) to account for artificial censoring due to deviation from the assigned treatment strategy. Because all patients are assigned to both treatment strategies at baseline, baseline confounding in treatment assignment is eliminated by design. Accordingly, adjustment for confounders is incorporated into models for the artificial censoring process rather than treatment assignment or outcome models. Follow-up will be structured in one-week discrete time intervals, and models will include flexible functions of time to allow for non-constant baseline hazards.

IPCW models will include key confounders selected based on clinical relevance and prior literature. Covariates common to all cancer types will include age, sex, race, ethnicity, ECOG performance status,[72] smoking status, practice setting (academic versus community),[73] insurance type,[74] and initial stage at diagnosis, all of which may influence both treatment duration decisions and outcomes. Separate IPCW models will be constructed for each cancer type to allow for confounders unique to each cancer. Within IPCW models for each cancer type, we will include cancer-specific confounders to account for differences in disease biology and treatment selection. For NSCLC, these include PD-L1 expression level,[75] treatment type (ICI alone or ICI plus chemotherapy) histology (squamous versus non-squamous), and key genomic alterations (e.g., KRAS, STK11, EGFR).[76,77] For bladder cancer, we will include treatment regimen (ICI alone or ICI plus enfortumab vedotin).[4] For kidney cancer, we will include IMDC (International Metastatic Renal Cell Carcinoma Database Consortium) risk score (poor, intermediate, favorable)[78] and treatment type (ICI alone, ICI doublet, or ICI plus targeted therapy). These variables collectively capture key factors influencing both treatment duration decisions and outcomes. Additional variables will be included as available in exploratory analyses, including the tumor biomarkers TMB or microsatellite instability status,[79] area-level socioeconomic status (Yost index),[80] and site of metastases (e.g., central nervous system involvement).[81,82]

From fitted pooled logistic regression models, we will estimate standardized survival curves under each treatment strategy and use these to calculate the primary estimand, difference in the probability of overall survival at two years. As a secondary estimand, we will also estimate the difference in restricted mean survival time (RMST) over the two-year follow-up period.

Non-inferiority will be assessed by comparing the difference in overall survival at two years between the continuation and discontinuation arms using the prespecified noninferiority margin of 5%. Specifically, discontinuation will be considered non-inferior to continuation if the upper bound of a one-sided 97.5% confidence interval for the difference in overall survival at two years is below 5%.

Exploratory analyses will include varying the discontinuation timepoint to explore discontinuation at 18 months or one year. Analyses will be conducted parallel to those described above for the two-year decision milestone. Exploratory subgroup analyses will evaluate heterogeneity of treatment effects (HTE) across factors including cancer type, age group, BMI, ECOG performance status, PD-L1 and/or TMB status, and calendar period of ICI initiation by including interaction terms between treatment strategy and the HTE factor of interest in pooled logistic regression models. To the extent that biomarkers such as PD-L1 are missing for certain cancers, we will limit secondary analyses to cancer types with low levels of missing data.

***Aim 2.*** We will use an approach parallel to Aim 1 to evaluate the effect of discontinuation of ICIs at two years versus continuation on irAEs among patients who are progression-free at the two-year milestone. Treatment strategies, eligibility criteria, start of follow-up, and implementation of the clone-censor-weight approach will be defined as described for Aim 1.

We will first examine the composite outcome of time to first occurrence of any of the irAEs under study (colitis, pneumonitis, hypothyroidism, adrenal insufficiency, or hypophysitis), followed by analyses of time to first occurrence of specific irAE types. Because death may preclude the occurrence of irAEs, it will be treated as a competing event. We will conduct descriptive analyses to summarize the frequency and timing of irAE events, including variation across cancer types and patient subgroups.

We will estimate the effect of treatment strategy on irAE outcomes using pooled logistic regression with IPCW, structured in discrete time intervals as described in Aim 1. Competing risks will be addressed by modeling cause-specific hazards for irAE and death and using these models to estimate standardized cumulative incidence functions under each treatment strategy. The primary estimand will be the difference in cumulative incidence of irAE at two years. Analyses of specific irAE types will follow the same approach.

***Aim 3.*** Building on the target trial emulation framework used in Aims 1 and 2, we will develop individualized risk prediction models for each of the two outcomes of interest (overall survival and irAEs) under each treatment strategy using regularized regression. In contrast to Aims 1 and 2, this aim seeks to provide individualized risk predictions as opposed to supporting causal inference. While there are several existing risk models for overall survival in advanced cancer patients,[83] these models focus on the setting of palliative care. No existing model specifically informs individualized risk prediction for patients who have been on treatment and progression-free for an extended period of time. Regularized regression improves prediction and reduces overfitting by adding a penalty term to the model-fitting process that shrinks regression coefficients toward zero. In prior work involving similar sample sizes and predictor dimensionality, our team has found that regularized regression performs as well as or better than more complex machine learning approaches while providing greater interpretability and stability.[84,85] We will develop prediction models in the full analytic cohort using the same pooled logistic regression approach leveraged in Aims 1 and 2, adding an elastic net penalty to stabilize estimation, with tuning parameters selected by cross-validation. Models will include a binary indicator for continuation versus discontinuation at two years, patient and tumor characteristics available at that timepoint, and the interaction between continuation/discontinuation and these characteristics. Characteristics included in risk model development will be: cancer type, age, sex, race, ethnicity, ECOG performance status, smoking status, practice setting (academic versus community), insurance type, initial stage at diagnosis, PD-L1 expression level, treatment type (ICI alone or in combination [another ICI, chemotherapy, or targeted therapy]), histology, genomic alterations (e.g., KRAS, STK11), IMDC risk score, TMB, microsatellite instability status, Yost index, and site of metastasis. For risk factors only relevant to a subset of cancer types (e.g. PD-L1 status, IMDC risk score) we will include interactions between the risk factor and relevant cancer type to allow these variables to contribute to risk estimation for that cancer type only. This modeling strategy will allow estimation of individualized predicted risks of overall survival and irAEs under each treatment strategy conditional on a patient's unique constellation of clinical and tumor characteristics. Given the limited sample size for some characteristics of interest, the use of the elastic net penalty allows us to stabilize parameter estimates by borrowing information across patient subgroups while also eliminating some terms from the model entirely when there is limited evidence to support an impact of the term on predictive performance. Model performance will be assessed using optimism-corrected internal validation via bootstrap resampling,[86] evaluating discrimination and calibration. Performance will be evaluated based on the bootstrap-corrected time-varying c-statistic, expected-to-observed ratio, calibration slope, and calibration intercept, all assessed at two years after the two-year decision milestone.

Following development and internal validation of the overall survival and irAE clinical risk prediction models, we will develop a user-friendly interactive risk prediction tool using R Shiny, an open-source software package. Our team has previously developed an R Shiny app for a clinical risk prediction model to inform decisions about surveillance breast imaging for breast cancer survivors. The first step in development of the tool will be adapting the final models into backend functions which will be accessed by the R Shiny app to calculate overall survival probability and cumulative incidence of irAEs at 1 and 2 years after the decision timepoint. Separate functions will be programmed for overall survival and irAE risk prediction under each treatment strategy (discontinuation versus continuation). We will verify the accuracy and calibration of output from these functions relative to the underlying risk prediction models to ensure fidelity of the software. The second step in tool development will consist of programming the user interface using R Shiny. A user-friendly results panel will provide numerical and graphical depictions of individualized predicted risks under each treatment strategy to support shared decision making. Under the guidance of the study team and using user-centered design principles, we will incorporate accessibility features including color contrast, inline form-field validation, placeholder examples, dynamic updating of output, and inline help. Once a prototype has been developed, we will seek feedback from patient and clinician members of our advisory board (see C.9) to ensure usability, interpretability, and clinical relevance. We expect future studies to further refine the prototype and test strategies to integrate it into clinical workflow and assess its usability and effectiveness.

We anticipate that prediction models will achieve good discriminatory accuracy based on the inclusion of established prognostic factors such as age, cancer type, stage at diagnosis, ECOG performance status, and tumor characteristics. However, the utility of the proposed decision support tool does not depend solely on predictive accuracy. Even if differences in outcomes between continuation and discontinuation are modest, individualized estimates of expected survival and irAE risk under each treatment strategy may provide valuable information on the balance of benefits and harms. We will seek feedback from our external advisory board regarding levels of model performance and presentation formats that are considered clinically useful. In addition, we will evaluate model performance across patient subgroups and examine the relative importance of prognostic factors to better understand characteristics that are the strongest predictors of outcomes.

**C.7. Missing data.** Based on preliminary analyses and prior experience with Flatiron data, we expect missingness in some confounder variables including ECOG performance status. As we have done in our prior studies using the Flatiron database, missingness will be addressed using multiple imputation under a missing at random assumption, with imputation models including all variables used in the analysis as well as outcomes and indicators of artificial censoring. We have previously conducted methodological investigations of the effectiveness of multiple imputation at mitigating bias due to missing confounder data in the Flatiron Health database and found this approach to be robust under a wide range of missing data mechanisms.[61,62] The number of imputations to be performed will be determined based on the fraction missing information, as recommended in the literature.[87] Results will be combined across imputations using Rubin's rules. Sensitivity analyses will use a tipping point approach to assess the robustness of results to departures from the missing at random assumption.

**C.8. Sample Size and Power.** Based on preliminary data from Flatiron Health, we estimate a total sample size of 7,837 patients (5,980 lung, 624 bladder, 589 kidney, and 644 melanoma) who initiated treatment with an ICI and were progression-free at two years. Of these patients, 2,405 discontinued treatment at the two-year decision point, while 5,432 continued treatment beyond two years. The study is powered to evaluate non-inferiority of discontinuation versus continuation with respect to overall survival. We define non-inferiority as an absolute difference in overall survival probability at two years of 5%, based on clinical judgement regarding acceptable tradeoffs between survival, toxicity, and treatment burden. With this sample size, we will have 91% power to demonstrate non-inferiority at a one-sided alpha of 0.025. Hypothesis testing for irAEs will follow superiority principles, testing the hypothesis that discontinuation results in lower cumulative incidence of irAEs compared to continuation. We will have 80% power to detect a 3.3% difference in cumulative incidence of irAEs at two-years at the two-sided alpha = 0.05 level. Final sample size and power estimates will be based on updated cohort counts from Flatiron and stakeholder-defined thresholds for clinical meaningfulness.

**C.9. Engagement and Dissemination.** We have engaged key stakeholders during proposal development, including patients receiving ICIs, their caregivers, and oncologists. In addition, we have formed an External Advisory Board (EAB, see letters) who will provide input throughout the study and support dissemination of our findings. Our EAB includes patient advocates with lived experience, academic clinicians (including NCCN guideline committee members), a community oncologist, outcomes researchers and methodologists, and a payer representative. This group will meet twice per year during the study period via video conferencing to provide input on select research design decisions, interpretation of study findings, and dissemination activities.

In Years 1 and 2, we will engage stakeholders through two one-hour virtual EAB meetings (two meetings per year, including one hour of meeting preparation) to elicit focused feedback on specific design decisions (e.g., choice of noninferiority margin, duration decision points for sensitivity analysis, refinement of irAE definitions). In Year 3, we will finalize analyses and develop a web-based prototype (an R Shiny app) that provides individualized estimates of survival and irAE risk under each treatment strategy. We will host two one-hour virtual EAB meetings (including one hour of meeting preparation) focused on a review of our final results, discussion of planned dissemination activities, review of dissemination materials, and charting a path for future work and next steps. We will hold an additional two one-hour virtual meetings (one for clinicians and one for patients) to obtain feedback on the user interface and display of information on the R Shiny prototype that will support future studies aimed at tool refinement and formal user testing.

Our findings will be disseminated through peer-reviewed publications, national meetings (e.g., ASCO), webinars to patient advocacy groups, and tailored one-page issue briefs for patients, clinicians, and payers. The EAB will assist in refining materials and disseminating these products. A summary of dissemination activities and factors influencing adoption of our findings for practice can be found in **Table 3**.

**Table 3.** Summary of end users, dissemination strategies, and factors influencing adoption of evidence.

| Activity / Issue | End User Groups | | |
|---|---|---|---|
| | **Patients & Caregivers** | **Oncology Providers** | **Payers** |
| Key dissemination activities | • Webinars with advocacy groups (Bladder Cancer Advocacy Network, Judith Nicholson Kidney Cancer Foundation, LUNGevity, Melanoma Research Alliance)<br>• 1-page patient summaries | • Publications & presentations<br>• 1-page clinical briefs | • 1-page policy briefs on real-world evidence |
| Barriers to uptake | • Misunderstanding of real-world evidence<br>• Fear or burden-driven preferences | • Reliance on anecdote<br>• Misunderstanding of real-world evidence | • Preference for early discontinuation regardless of outcome |
| Strategies to address barriers | • Advocacy-based education (webinars)<br>• Clear patient materials | • Transparent reporting (TARGET methodology)<br>• Clinically focused summaries | • Evidence aligned with coverage decisions |
| Facilitators to adoption | • Real-world populations<br>• Relevant outcomes | • Methodologic rigor<br>• Clinical relevance | • Relevance to value-based care |

**C.10. Impact.** This study addresses a critical and unresolved decision faced by a variety of stakeholders.

- *For patients and caregivers*: Provides evidence-based guidance on ICI duration and personalized estimates of risks, reducing uncertainty.
- *For clinicians*: Informs a high-stakes decision currently guided by convention, with both population-level and individualized estimates to support shared decision-making.
- *For health systems and payers*: Clarifies whether continued ICI therapy beyond two years provides meaningful benefit, informing coverage decisions and optimizing the use of costly treatments.
- *For population health*: Reduces practice variation and improves consistency of care through dissemination of rigorous real-world evidence.
- *For researchers*: Demonstrates how TTE can address pressing noninferiority questions and generate clinically actionable evidence.

We therefore expect this study to fill a major evidence gap in cancer care by providing much needed rigorous real-world evidence on the optimal duration of ICI therapy in patients with advanced cancer. By integrating comparative effectiveness analyses with individualized risk prediction, this work will support more informed, personalized treatment decisions while potentially reducing unnecessary toxicity and improving the value of cancer care. More broadly, this work will demonstrate how modern causal inference methods applied to real-world data can address clinically important treatment-duration questions that are unlikely to be resolved through randomized trials alone.

## REFERENCES


1. American Cancer Society. *Cancer Facts & Figures 2025*. 2025. https://www.cancer.org/content/dam/cancer-org/research/cancer-facts-and-statistics/annual-cancer-facts-and-figures/2025/2025-cancer-facts-and-figures-acs.pdf

2. Robert C, Carlino MS, McNeil C, et al. Seven-Year Follow-Up of the Phase III KEYNOTE-006 Study: Pembrolizumab Versus Ipilimumab in Advanced Melanoma. *J Clin Oncol Off J Am Soc Clin Oncol*. 2023;41(24):3998-4003. doi:10.1200/JCO.22.01599

3. Motzer R, Alekseev B, Rha SY, et al. Lenvatinib plus Pembrolizumab or Everolimus for Advanced Renal Cell Carcinoma. *N Engl J Med*. 2021;384(14):1289-1300. doi:10.1056/NEJMoa2035716

4. Powles T, Valderrama BP, Gupta S, et al. Enfortumab Vedotin and Pembrolizumab in Untreated Advanced Urothelial Cancer. *N Engl J Med*. 2024;390(10):875-888. doi:10.1056/NEJMoa2312117

5. Garassino MC, Gadgeel S, Speranza G, et al. Pembrolizumab Plus Pemetrexed and Platinum in Nonsquamous Non-Small-Cell Lung Cancer: 5-Year Outcomes From the Phase 3 KEYNOTE-189 Study. *J Clin Oncol Off J Am Soc Clin Oncol*. 2023;41(11):1992-1998. doi:10.1200/JCO.22.01989

6. Voruganti T, Soulos PR, Mamtani R, Presley CJ, Gross CP. Association Between Age and Survival Trends in Advanced Non-Small Cell Lung Cancer After Adoption of Immunotherapy. *JAMA Oncol*. 2023;9(3):334-341. doi:10.1001/jamaoncol.2022.6901

7. Parikh RB, Min EJ, Wileyto EP, et al. Uptake and Survival Outcomes Following Immune Checkpoint Inhibitor Therapy Among Trial-Ineligible Patients With Advanced Solid Cancers. *JAMA Oncol*. 2021;7(12):1843-1850. doi:10.1001/jamaoncol.2021.4971

8. Sun L, Bleiberg B, Hwang WT, et al. Association Between Duration of Immunotherapy and Overall Survival in Advanced Non-Small Cell Lung Cancer. *JAMA Oncol*. 2023;9(8):1075-1082. doi:10.1001/jamaoncol.2023.1891

9. Postow MA, Sidlow R, Hellmann MD. Immune-Related Adverse Events Associated with Immune Checkpoint Blockade. *N Engl J Med*. 2018;378(2):158-168. doi:10.1056/NEJMra1703481

10. Aguiar-Ibáñez R, Scherrer E, Grebennik D, et al. Time and productivity loss associated with immunotherapy infusions for the treatment of melanoma in the United States: a survey of health care professionals and patients. *BMC Health Serv Res*. 2023;23(1):136. doi:10.1186/s12913-022-08904-4

11. Candido W, Eggen AC, Jalving M, et al. Quality of life, neurocognitive functioning, psychological issues, sexuality and comorbidity more than 2 years after commencing immune checkpoint inhibitor treatment. *J Immunother Cancer*. 2025;13(3):e011168. doi:10.1136/jitc-2024-011168

12. Pardoll DM. The blockade of immune checkpoints in cancer immunotherapy. *Nat Rev Cancer*. 2012;12(4):252-264. doi:10.1038/nrc3239

13. Gandhi L, Rodríguez-Abreu D, Gadgeel S, et al. Pembrolizumab plus Chemotherapy in Metastatic Non-Small-Cell Lung Cancer. *N Engl J Med*. 2018;378(22):2078-2092. doi:10.1056/NEJMoa1801005

14. Waterhouse DM, Garon EB, Chandler J, et al. Continuous Versus 1-Year Fixed-Duration Nivolumab in Previously Treated Advanced Non-Small-Cell Lung Cancer: CheckMate 153. *J Clin Oncol Off J Am Soc Clin Oncol*. 2020;38(33):3863-3873. doi:10.1200/JCO.20.00131

15. Coen O, Corrie P, Marshall H, et al. The DANTE trial protocol: a randomised phase III trial to evaluate the Duration of ANti-PD-1 monoclonal antibody Treatment in patients with metastatic mElanoma. *BMC Cancer*. 2021;21(1):761. doi:10.1186/s12885-021-08509-w

16. Gemelli M, Sala L, Morabito A, et al. Real-world outcomes of immunotherapy discontinuation at two years in advanced non-small cell lung cancer: Final results of the I-STOP study. *Eur J Cancer*. 2026;237:116579. doi:10.1016/j.ejca.2026.116579

17. Pandey T, Bhatti SA. Efficacy and safety of two-year fixed versus continuous immune checkpoint inhibitor therapy in advanced or metastatic non-small cell lung cancer: a systematic review. *Cancer Immunol Immunother CII*. 2025;74(10):317. doi:10.1007/s00262-025-04143-8

18. Bogani G, Cinquini M, Signorelli D, et al. A systematic review and meta-analysis on the optimal treatment duration of checkpoint inhibitoRS in solid tumors: The OTHERS study. *Crit Rev Oncol Hematol*. 2023;187:104016. doi:10.1016/j.critrevonc.2023.104016

19. Perez L, Samlowski W, Lopez-Flores R. Outcome of Elective Checkpoint Inhibitor Discontinuation in Patients with Metastatic Melanoma Who Achieved a Complete Remission: Real-World Data. *Biomedicines*. 2022;10(5):1144. doi:10.3390/biomedicines10051144

20. Martini DJ, Hamieh L, McKay RR, et al. Durable Clinical Benefit in Metastatic Renal Cell Carcinoma Patients Who Discontinue PD-1/PD-L1 Therapy for Immune-Related Adverse Events. *Cancer Immunol Res*. 2018;6(4):402-408. doi:10.1158/2326-6066.CIR-17-0220

21. Amiot M, Mortier L, Dalle S, et al. When to stop immunotherapy for advanced melanoma: the emulated target trials. *EClinicalMedicine*. 2024;78:102960. doi:10.1016/j.eclinm.2024.102960

22. Remon J, Bortolot M, Bironzo P, et al. De-Escalation Strategies With Immune Checkpoint Blockers in Non-Small Cell Lung Cancer: Do We Already Have Enough Evidence? *J Clin Oncol Off J Am Soc Clin Oncol*. 2025;43(9):1148-1156. doi:10.1200/JCO-24-02347

23. Bazarbachi AH, Magrini N, Aziz Z, Fojo T. Evidence for a reduction in number of cycles of immune checkpoint inhibitors. *Lancet Oncol*. 2025;26(1):9-11. doi:10.1016/S1470-2045(24)00594-1

24. Davies MA. Is It Safe to Stop Anti-PD-1 Immunotherapy in Patients With Metastatic Melanoma Who Achieve a Complete Response? *J Clin Oncol Off J Am Soc Clin Oncol*. 2020;38(15):1645-1647. doi:10.1200/JCO.20.00136

25. Merrick S, Rush HL, Daniels S, et al. Hit it hard: qualitative patient perspectives on the optimisation of immune checkpoint inhibition. *Br J Cancer*. 2024;131(3):515-523. doi:10.1038/s41416-024-02756-x

26. Chubak J, Buist DSM, Boudreau DM, Rossing MA, Lumley T, Weiss NS. Breast cancer recurrence risk in relation to antidepressant use after diagnosis. *Breast Cancer Res Treat*. 2008;112(1):123-132. doi:10.1007/s10549-007-9828-9

27. Chubak J, Bowles EJ, Yu O, Buist DS, Fujii M, Boudreau DM. Breast cancer recurrence in relation to antidepressant use. *Cancer Causes Control*. 2016;27(1):125-136. doi:10.1007/s10552-015-0689-y

28. Chubak J, Yu O, Ziebell RA, et al. Risk of colon cancer recurrence in relation to diabetes. *Cancer Causes Control CCC*. 2018;29(11):1093-1103. doi:10.1007/s10552-018-1083-3

29. Pocobelli G, Yu O, Ziebell RA, et al. Use of antidepressants after colon cancer diagnosis and risk of recurrence. *Psychooncology*. 2019;28(4):750-758. doi:10.1002/pon.5015

30. Bowles EJA, Yu O, Ziebell R, et al. Cardiovascular medication use and risks of colon cancer recurrences and additional cancer events: a cohort study. *BMC Cancer*. 2019;19(1):270. doi:10.1186/s12885-019-5493-8

31. Sarma EA, Thompson MJ, Bowles EJA, et al. Patient and tumour characteristics of screening-age adults diagnosed with screen-detected versus symptomatic colon cancer. *Colorectal Dis Off J Assoc Coloproctology G B Irel*. 2022;24(11):1344-1351. doi:10.1111/codi.16232

32. Onega T, Weiss J, Kerlikowske K, et al. The influence of race/ethnicity and place of service on breast reconstruction for Medicare beneficiaries with mastectomy. *SpringerPlus*. 2014;3(1):416. doi:10.1186/2193-1801-3-416

33. Onega T, Tosteson ANA, Weiss J, et al. Costs of diagnostic and preoperative workup with and without breast MRI in older women with a breast cancer diagnosis. *BMC Health Serv Res*. 2016;16(1):76. doi:10.1186/s12913-016-1317-6

34. Onega T, Weiss JE, Buist DS, et al. Breast MRI in the Diagnostic and Preoperative Workup among Medicare Beneficiaries with Breast Cancer. *Med Care*. 2016;54(7):719-724. doi:10.1097/MLR.0000000000000542

35. Feld E, Harton J, Meropol NJ, et al. Effectiveness of First-line Immune Checkpoint Blockade Versus Carboplatin-based Chemotherapy for Metastatic Urothelial Cancer. *Eur Urol*. 2019;76(4):524-532. doi:10.1016/j.eururo.2019.07.032

36. Parikh RB, Takvorian SU, Vader D, et al. Impact of the COVID-19 Pandemic on Treatment Patterns for Patients With Metastatic Solid Cancer in the United States. *JNCI J Natl Cancer Inst*. 2022;114(4):571-578. doi:10.1093/jnci/djab225

37. Parikh RB, Hubbard RA, Wang E, et al. Exposure to US Cancer Drugs With Lack of Confirmed Benefit After US Food and Drug Administration Accelerated Approval. *JAMA Oncol*. 2023;9(4):567-569. doi:10.1001/jamaoncol.2022.7770

38. Vader DT, Parikh RB, Li H, Imai K, Hubbard RA, Mamtani R. Impact of Label Restriction on Checkpoint-Inhibitor Use in Bladder Cancer and Changes in Mortality. *JNCI Cancer Spectr*. 2022;6(4):pkac050. doi:10.1093/jncics/pkac050

39. Nimgaonkar V, Hubbard RA, Carpenter EL, Mamtani R. Biomarker Testing, Treatment Uptake, and Survival Among Patients With Urothelial Cancer Receiving Gene-Targeted Therapy. *JAMA Oncol*. 2022;8(7):1070-1072. doi:10.1001/jamaoncol.2022.1167

40. Shen J, Linn KA, Clark AS, Mamtani R, Hubbard RA. Missingness in Eligibility Criteria for Target Trial Emulation in EHR With Survival Outcomes. *Stat Med*. 2026;45(8-9):e70500. doi:10.1002/sim.70500

41. Tannir NM, Albigès L, McDermott DF, et al. Nivolumab plus ipilimumab versus sunitinib for first-line treatment of advanced renal cell carcinoma: extended 8-year follow-up results of efficacy and safety from the phase III CheckMate 214 trial. *Ann Oncol Off J Eur Soc Med Oncol*. 2024;35(11):1026-1038. doi:10.1016/j.annonc.2024.07.727

42. Jassem J, de Marinis F, Giaccone G, et al. Updated Overall Survival Analysis From IMpower110: Atezolizumab Versus Platinum-Based Chemotherapy in Treatment-Naive Programmed Death-Ligand 1-Selected NSCLC. *J Thorac Oncol Off Publ Int Assoc Study Lung Cancer*. 2021;16(11):1872-1882. doi:10.1016/j.jtho.2021.06.019

43. West H, McCleod M, Hussein M, et al. Atezolizumab in combination with carboplatin plus nab-paclitaxel chemotherapy compared with chemotherapy alone as first-line treatment for metastatic non-squamous non-small-cell lung cancer (IMpower130): a multicentre, randomised, open-label, phase 3 trial. *Lancet Oncol*. 2019;20(7):924-937. doi:10.1016/S1470-2045(19)30167-6

44. Danaei, G, Rasouli, B, Chubak, J. *Developing New Methods for Comparing Treatments in Case-Control Studies*. 2017. Accessed September 13, 2023. https://doi.org/10.25302/07.2021.ME.160936748

45. Duchesneau ED, Jackson BE, Webster-Clark M, et al. The Timing, the Treatment, the Question: Comparison of Epidemiologic Approaches to Minimize Immortal Time Bias in Real-World Data Using a Surgical Oncology Example. *Cancer Epidemiol Biomark Prev Publ Am Assoc Cancer Res Cosponsored Am Soc Prev Oncol*. 2022;31(11):2079-2086. doi:10.1158/1055-9965.EPI-22-0495

46. Gaber CE, Shaheen NJ, Edwards JK, et al. Trimodality Therapy vs Definitive Chemoradiation in Older Adults With Locally Advanced Esophageal Cancer. *JNCI Cancer Spectr*. 2022;6(6):pkac069. doi:10.1093/jncics/pkac069

47. Su IH, Lund JL, Gaber CE, Sanoff HK, Strassle PD, Duchesneau ED. Regarding “Neoadjuvant Versus Adjuvant Chemotherapy for Resectable Metastatic Colon Cancer in Non-academic and Academic Programs.” *The Oncologist*. 2023;28(7):e588-e589. doi:10.1093/oncolo/oyad130

48. Gaber CE, Ghazarian AA, Strassle PD, et al. De-Mystifying the Clone-Censor-Weight Method for Causal Research Using Observational Data: A Primer for Cancer Researchers. *Cancer Med*. 2024;13(23):e70461. doi:10.1002/cam4.70461

49. Sharma P, Goswami S, Raychaudhuri D, et al. Immune checkpoint therapy-current perspectives and future directions. *Cell*. 2023;186(8):1652-1669. doi:10.1016/j.cell.2023.03.006

50. Owen CN, Bai X, Quah T, et al. Delayed immune-related adverse events with anti-PD-1-based immunotherapy in melanoma. *Ann Oncol Off J Eur Soc Med Oncol*. 2021;32(7):917-925. doi:10.1016/j.annonc.2021.03.204

51. Brookhart MA. Counterpoint: The Treatment Decision Design. *Am J Epidemiol*. 2015;182(10):840-845. doi:10.1093/aje/kwv214

52. van der Heijden MS, Sonpavde G, Powles T, et al. Nivolumab plus Gemcitabine-Cisplatin in Advanced Urothelial Carcinoma. *N Engl J Med*. 2023;389(19):1778-1789. doi:10.1056/NEJMoa2309863

53. Rini BI, Plimack ER, Stus V, et al. Pembrolizumab plus Axitinib versus Sunitinib for Advanced Renal-Cell Carcinoma. *N Engl J Med*. 2019;380(12):1116-1127. doi:10.1056/NEJMoa1816714

54. Choueiri TK, Powles T, Burotto M, et al. Nivolumab plus Cabozantinib versus Sunitinib for Advanced Renal-Cell Carcinoma. *N Engl J Med*. 2021;384(9):829-841. doi:10.1056/NEJMoa2026982

55. Paz-Ares L, Luft A, Vicente D, et al. Pembrolizumab plus Chemotherapy for Squamous Non-Small-Cell Lung Cancer. *N Engl J Med*. 2018;379(21):2040-2051. doi:10.1056/NEJMoa1810865

56. Makharadze T, Gogishvili M, Melkadze T, et al. Cemiplimab Plus Chemotherapy Versus Chemotherapy Alone in Advanced NSCLC: 2-Year Follow-Up From the Phase 3 EMPOWER-Lung 3 Part 2 Trial. *J Thorac Oncol Off Publ Int Assoc Study Lung Cancer*. 2023;18(6):755-768. doi:10.1016/j.jtho.2023.03.008

57. Mok TSK, Wu YL, Kudaba I, et al. Pembrolizumab versus chemotherapy for previously untreated, PD-L1-expressing, locally advanced or metastatic non-small-cell lung cancer (KEYNOTE-042): a randomised, open-label, controlled, phase 3 trial. *Lancet*. 2019;393(10183):1819-1830. doi:10.1016/S0140-6736(18)32409-7

58. Kilickap S, Baramidze A, Sezer A, et al. Cemiplimab Monotherapy for First-Line Treatment of Patients with Advanced NSCLC With PD-L1 Expression of 50% or Higher: Five-Year Outcomes of EMPOWER-Lung 1. *J Thorac Oncol Off Publ Int Assoc Study Lung Cancer*. 2025;20(7):941-954. doi:10.1016/j.jtho.2025.03.033

59. Reck M, Rodríguez-Abreu D, Robinson AG, et al. Pembrolizumab versus Chemotherapy for PD-L1-Positive Non-Small-Cell Lung Cancer. *N Engl J Med*. 2016;375(19):1823-1833. doi:10.1056/NEJMoa1606774

60. Decruyenaere A, Christine G, Sylvie R, et al. Optimal treatment duration in metastatic renal cell carcinoma patients responding to immune checkpoint inhibitors: should we treat beyond two years? *Acta Oncol*. 2025;64:979-988. doi:10.2340/1651-226X.2025.43876

61. Martin T, Krishnan A, Yong K, et al. Comparative effectiveness of ciltacabtagene autoleucel in CARTITUDE-1 versus physician’s choice of therapy in the Flatiron Health multiple myeloma cohort registry

for the treatment of patients with relapsed or refractory multiple myeloma. *eJHaem*. 2022;3(1):97-108. doi:10.1002/jha2.312

62. DeMichele A, Cristofanilli M, Brufsky A, et al. Comparative effectiveness of first-line palbociclib plus letrozole versus letrozole alone for HR+/HER2− metastatic breast cancer in US real-world clinical practice. *Breast Cancer Res*. 2021;23(1):37. doi:10.1186/s13058-021-01409-8

63. Bagley SJ, Talento S, Mitra N, et al. Comparative Effectiveness of Carboplatin/Pemetrexed With Versus Without Bevacizumab for Advanced Nonsquamous Non–Small Cell Lung Cancer. *J Natl Compr Canc Netw*. 2019;17(5):469-477. doi:10.6004/jnccn.2018.7102

64. Moser JC, Wei G, Colonna SV, Grossmann KF, Patel S, Hyngstrom JR. Comparative-effectiveness of pembrolizumab vs. nivolumab for patients with metastatic melanoma. *Acta Oncol*. 2020;59(4):434-437. doi:10.1080/0284186X.2020.1712473

65. Berger ML, Curtis MD, Smith G, Harnett J, Abernethy AP. Opportunities and challenges in leveraging electronic health record data in oncology. *Future Oncol*. 2016;12(10):1261-1274. doi:10.2217/fon-2015-0043

66. Curtis MD, Griffith SD, Tucker M, et al. Development and Validation of a High-Quality Composite Real-World Mortality Endpoint. *Health Serv Res*. 2018;53(6):4460-4476. doi:10.1111/1475-6773.12872

67. Griffith SD, Miksad RA, Calkins G, et al. Characterizing the Feasibility and Performance of Real-World Tumor Progression End Points and Their Association With Overall Survival in a Large Advanced Non–Small-Cell Lung Cancer Data Set. *JCO Clin Cancer Inform*. 2019;(3):1-13. doi:10.1200/CCI.19.00013

68. Abernethy AP, Gippetti J, Parulkar R, Revol C. Use of Electronic Health Record Data for Quality Reporting. *J Oncol Pract*. 2017;13(8):530-534. doi:10.1200/JOP.2017.024224

69. Miksad RA, Abernethy AP. Harnessing the Power of Real-World Evidence (RWE): A Checklist to Ensure Regulatory-Grade Data Quality. *Clin Pharmacol Ther*. 2018;103(2):202-205. doi:10.1002/cpt.946

70. Estevez M, Singh N, Dyson L, et al. Ensuring Reliability of Curated Electronic Health Record–Derived Data: The Validation of Accuracy for Large Language Model–/Machine Learning–Extracted Information and Data (VALID) Framework. *JCO Clin Cancer Inform*. 2026;10(1):e2500215. doi:10.1200/CCI-25-00215

71. Carrigan G, Whipple S, Taylor MD, et al. An evaluation of the impact of missing deaths on overall survival analyses of advanced non–small cell lung cancer patients conducted in an electronic health records database. *Pharmacoepidemiol Drug Saf*. 2019;28(5):572-581. doi:10.1002/pds.4758

72. Ma X, Long L, Moon S, Adamson BJS, Baxi SS. Comparison of Population Characteristics in Real-World Clinical Oncology Databases in the US: Flatiron Health, SEER, and NPCR. *medRxiv*. Preprint posted online June 7, 2023:2020.03.16.20037143. doi:10.1101/2020.03.16.20037143

73. Vader DT, Mamtani R, Li Y, Griffith SD, Calip GS, Hubbard RA. Inverse Probability of Treatment Weighting and Confounder Missingness in Electronic Health Record-based Analyses: A Comparison of Approaches Using Plasmode Simulation. *Epidemiology*. 2023;34(4):520. doi:10.1097/EDE.0000000000001618

74. Getz K, Hubbard RA, Linn KA. Performance of Multiple Imputation Using Modern Machine Learning Methods in Electronic Health Records Data. *Epidemiology*. 2022;34(2):206-215. doi:10.1097/EDE.0000000000001578

75. Hernan MA. Counterpoint: epidemiology to guide decision-making: moving away from practice-free research. *Am J Epidemiol*. 2015;182(10):834-839. doi:10.1093/aje/kwv215

76. Cashin AG, Hansford HJ, Hernán MA, et al. Transparent Reporting of Observational Studies Emulating a Target Trial—The TARGET Statement. *JAMA*. 2025;334(12):1084-1093. doi:10.1001/jama.2025.13350

77. Kuehne F, Arvandi M, Hess LM, et al. Causal analyses with target trial emulation for real-world evidence removed large self-inflicted biases: systematic bias assessment of ovarian cancer treatment effectiveness. *J Clin Epidemiol*. 2022;152:269-280. doi:10.1016/j.jclinepi.2022.10.005

78. Merola D. The Use of Oncology Electronic Health Record Databases to Assess the Effectiveness of Breast Cancer Treatment. *Harv Univ Grad Sch Arts Sci*. 2022;Doctoral dissertation.

79. Gaber CE, Hanson KA, Kim S, Lund JL, Lee TA, Murray EJ. The Clone-Censor-Weight Method in Pharmacoepidemiologic Research: Foundations and Methodological Implementation. *Curr Epidemiol Rep*. Published online February 17, 2024. doi:10.1007/s40471-024-00346-2

80. Cain LE, Robins JM, Lanoy E, Logan R, Costagliola D, Hernán MA. When to start treatment? A systematic approach to the comparison of dynamic regimes using observational data. *Int J Biostat*. 2010;6(2):Article 18. doi:10.2202/1557-4679.1212

81. West H (Jack), Jin JO. Performance Status in Patients With Cancer. *JAMA Oncol*. 2015;1(7):998. doi:10.1001/jamaoncol.2015.3113

82. Hui D, Hess K, dos Santos R, Chisholm G, Bruera E. A diagnostic model for impending death in cancer patients: Preliminary report. *Cancer*. 2015;121(21):3914-3921. doi:10.1002/cncr.29602

83. Hu JC, Nguyen P, Mao J, et al. Increase in Prostate Cancer Distant Metastases at Diagnosis in the United States. *JAMA Oncol*. 2017;3(5):705-707. doi:10.1001/jamaoncol.2016.5465

84. Skoulidis F, Goldberg ME, Greenawalt DM, et al. STK11/LKB1 Mutations and PD-1 Inhibitor Resistance in KRAS-Mutant Lung Adenocarcinoma. *Cancer Discov*. 2018;8(7):822-835. doi:10.1158/2159-8290.CD-18-0099

85. Mazieres J, Drilon A, Lusque A, et al. Immune checkpoint inhibitors for patients with advanced lung cancer and oncogenic driver alterations: results from the IMMUNOTARGET registry. *Ann Oncol Off J Eur Soc Med Oncol*. 2019;30(8):1321-1328. doi:10.1093/annonc/mdz167

86. Heng DYC, Xie W, Regan MM, et al. Prognostic factors for overall survival in patients with metastatic renal cell carcinoma treated with vascular endothelial growth factor-targeted agents: results from a large, multicenter study. *J Clin Oncol Off J Am Soc Clin Oncol*. 2009;27(34):5794-5799. doi:10.1200/JCO.2008.21.4809

87. Marabelle A, Fakih M, Lopez J, et al. Association of tumour mutational burden with outcomes in patients with advanced solid tumours treated with pembrolizumab: prospective biomarker analysis of the multicohort, open-label, phase 2 KEYNOTE-158 study. *Lancet Oncol*. 2020;21(10):1353-1365. doi:10.1016/S1470-2045(20)30445-9

88. Yost K, Perkins C, Cohen R, Morris C, Wright W. Socioeconomic status and breast cancer incidence in California for different race/ethnic groups. *Cancer Causes Control CCC*. 2001;12(8):703-711. doi:10.1023/a:1011240019516

89. Deng Y, Chi P, Lan P, et al. Neoadjuvant Modified FOLFOX6 With or Without Radiation Versus Fluorouracil Plus Radiation for Locally Advanced Rectal Cancer: Final Results of the Chinese FOWARC Trial. *J Clin Oncol Off J Am Soc Clin Oncol*. 2019;37(34):3223-3233. doi:10.1200/JCO.18.02309

90. Bajorin DF, Dodd PM, Mazumdar M, et al. Long-term survival in metastatic transitional-cell carcinoma and prognostic factors predicting outcome of therapy. *J Clin Oncol Off J Am Soc Clin Oncol*. 1999;17(10):3173-3181. doi:10.1200/JCO.1999.17.10.3173

91. Su YR, Buist DSM, Lee JM, et al. Performance of Statistical and Machine Learning Risk Prediction Models for Surveillance Benefits and Failures in Breast Cancer Survivors. *Cancer Epidemiol Biomarkers Prev*. 2023;32(4):561-571. doi:10.1158/1055-9965.EPI-22-0677

92. White IR, Royston P, Wood AM. Multiple imputation using chained equations: issues and guidance for practice. *Stat Med*. 2011;30(4):377-399.